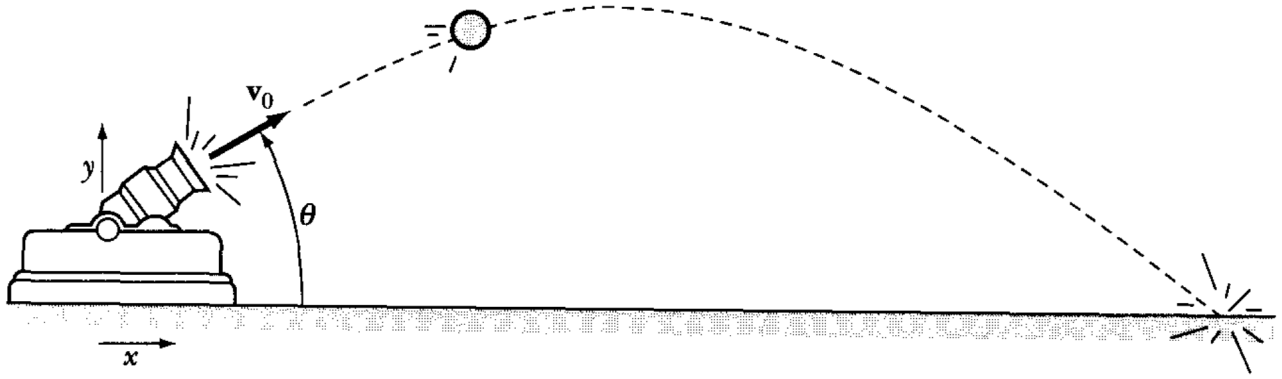
Advanced Classical Mechanics (PHYS 615) **Fall 2026**
Extra Credit Project 1 (due on date of Exam I)

The points obtained in this project will be added towards Exam I. To obtain full credit, your computer source code + calculations done on paper **MUST** be provided along with the plots.

Projectile Motion with Drag: **(5 pts)** Consider a projectile motion with drag force of the form $\mathbf{F} = -km\mathbf{v}$



Show that the projectile flight time and range are

$$T = \frac{kV + g}{gk}(1 - e^{-kT}) \tag{1a}$$

$$R = \frac{U}{k}(1 - e^{-kT}) \tag{1b}$$

where k is a drag coefficient and $V \equiv v_0 \sin\theta$ and $U \equiv v_0 \cos\theta$ are the initial y- and x-components of the projectile speed.

Note that the time of flight T is a transcendental equation and can't be solved analytically. Two methods for solving this problem are (i) *Perturbative* and (ii) *Numerical*

Perturbative Approach: **(5 pts)**

To use the perturbation method, we find an expansion parameter coupling constant that is normally small. In the present case, this parameter is the retarding force constant k, because we have already solved the present problem with $k = 0$ in the lecture, and now we would like to turn on the retarding force, but let k be small. We therefore expand the exponential term e^{-kT} in the flight time T (Eq. 1a) in a power series with the intention of keeping only the lowest terms of k^n, where k is our expansion parameter. Keep only terms in the expansion through k^3 and show it can be expressed as

$$T = \frac{2V/g}{1 + kV/g} + \frac{1}{3}kT^2 \tag{2}$$

Note that we now have the expansion parameter in the denominator of the first term, which we need to expand this term in a power series $1 / (1 + kV/g) \sim \ldots$ keeping only terms through k^2, since we only have terms through k in the T expression above. Now substitute the power series and keep only terms in k to first order to show that the flight time simplifies to

$$T = \frac{2V}{g} + \left(\frac{T^2}{3} - \frac{2V^2}{g^2}\right)k + O(k^2) \tag{3}$$

where we choose to neglect $O(k^2)$, the terms of k^2 and higher order.
Note from this expression, in the limit where $k \to 0$, the flight time reduces to the expected expression $T_0 = 2V/g$ for a projectile without air drag.

If k is small enough (but not vanishing), we can substitute the approximate expression $T \approx T_0 \equiv 2V_0/g$, into the right hand side of Eq. 3 to show that it yields the approximate flight time

$$T \approx \frac{2V}{g}\left(1 - \frac{kV}{3g}\right) \tag{4}$$

Now to obtain an approximate expression for the range, apply a Taylor expansion to the term e^{-kT} in the range (Eq.1b) using k as the *expansion parameter* and keeping terms only up k^2, to show it yields the expression

$$R \approx \frac{U}{k}\left(kT - \frac{1}{2}k^2T^2 + \frac{1}{6}k^3T^3 - \dots\right) \tag{5}$$

Now substitute the approximate expression for the flight time (Eq. 4) to the range (Eq. 5) keeping only up to k^2 terms, and re-arrange the equation to show that

$$R \approx R_0\left[1 - 4\left(\frac{kV}{3g}\right) + 9\left(\frac{kV}{3g}\right)^2\right] \tag{6}$$

where $R_0 \equiv 2UV/g$ (projectile range without drag)

Numerical Approach: **(10 pts)** To solve a transcendental equation, we use *Newtons's Method* numerical approach which is a root-finding algorithm, which has the recurring form:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

where x_{n+1} is the root after n-recursive steps. To use this method, the function must be of the form $f(x) = 0$ and it requires an initial guess of the root x that satisfies the equation. In our case, for a given value of k, we find the root T, and used the (k, T) pair as input into the range equation, (Eq. 1b) .

Solve the transcendental equation using Newton's Method.

step 1: need to put (Eq. 1a) in the form: $f(k, T) = 0$

step 2: for an arbitrary value of $0 < k \leq 0.08$, find the root T, as follows

$$T_{n+1} = T_n - \frac{f(k, T_n)}{\mathrm{d}f(k, T_n)/dT}$$

Use the initial conditions to get the iteration process going, and try using *n=100* iterations

initial conditions *(n=0)*:
$v_0 = 600$ m/s
$\theta = 60^\circ$
$T_0 = 150$ sec (initial guess will not necessarily satisfy $f(k, T_0) = 0$)

after n-iterations, the algorithm should have found the root that satisfies the equation $f(k, T) = 0$,

step 3: now that you have a (k_i, T_i) pair for an i-th arbitrary value of k, use it as input into the range equation, $R_i(k_i, T_i)$

step 4: at this point, you should have a procedure established for determining the values (k_i, T_i, R_i); now repeat the procedure for a range of k-values from 0.0000001 to 0.08 with step-size of $dk = 0.001$; ** note: since k is in the denominator, $k \neq 0$

step 5: make a plot of R vs. k to examine how the projectile range varies with the drag coefficient ** note: negative ranges are unphysical, therefore, when making the plots, set the lower limit on the y-axis to zero.
Add proper labels and a legend to each of the plots.

Now, to check the *perturbative* versus the *numerical* method, make a plot overlay of:

i) R vs. k (*perturbative* **linear** approximation) [keep only linear term in (6)]
ii) R vs. k (*perturbative* **quadratic** approximation) [linear term + quadratic]
iii) R vs. k (**numerical** approximation)

Add proper labels and a legend to each of the plots.

Write a brief paragraph commenting on your results. Is it what you expected to happen ? Elaborate.